Governance Signals for Cooperation Stability Under Adversarial Strategy Injection in Sequential Fishery Commons

Ameer Alhashemi

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Abstract

We study cooperation robustness in a minimal sequential fishery commons under adversarial strategy injection. Rather than asking whether cooperation emerges in a fixed population, we evaluate whether governance signals preserve cooperation under repeated strategy turnover and held-out regime shifts. The paper_v1 cycle uses a matched matrix across two regime tiers (easy_v1 and medium_v1), two injector types (mutation and live JSON-based LLM injection via Ollama), and three governance conditions (none, monitoring, and monitoring with sanctions). In medium_v1 high-power reruns, monitoring with sanctions reduces held-out collapse relative to none for both injectors with positive confidence bounds, supporting invasion-stability gains from minimal governance. Hard-tier results are treated as appendix-only stress evidence due to saturation.

1 Introduction

Classic commons theory predicts over-extraction under open access and highlights institutional design as the key lever for sustainable outcomes [2, 6, 10]. In machine learning, sequential social dilemma research and benchmark design emphasize policy-level dynamics, long horizons, and held-out evaluation rather than one-shot game payoffs [1, 3, 4]. Recent LLM-agent commons work has shown that cooperation can emerge in simulated societies, but robustness under distribution shift and repeated adversarial pressure remains weakly characterized [7, 8].

This work studies governance robustness in a minimal sequential fishery commons under adversarial strategy injection. The core research question is: *Which governance signals stabilize cooperation under adversarial strategy injection, and does this hold across regime difficulty and injector type, comparing mutation against live LLM generation?*

The key gap is a missing *invasion-stability protocol* for commons governance. Prior commons simulations typically evaluate emergence in mostly fixed populations, while continuous adversarial generation pressure has been explored mainly in other domains, such as program-level Red Queen dynamics [9]. Broader policy discussions around generative-agent societies also stress realism and evaluation discipline but do not provide this protocol for renewable commons governance [5]. This leaves an open problem: measuring whether governance signals remain effective when new exploitative strategies are repeatedly generated and injected.

The evaluation is designed as a matched comparison rather than a one-off demonstration. We use the same experiment budget and protocol across governance conditions, perform selection in train regimes, and evaluate robustness in held-out regimes. Outcomes are reported with uncertainty intervals from run-level summaries.

The cycle contribution is an evidence package for invasion-stability rather than a new environment feature. Concretely, we provide a matched governance matrix over two benchmark tiers, easy_v1

and medium_v1, and two injector types, mutation and live JSON-based LLM injection. We report confidence intervals for collapse reduction and supporting robustness metrics, provide a composite governance ranking that aggregates collapse, stock, and survival effects, and apply explicit gate logic that determines whether claims are supported or invalidated by regime saturation.

Hard-tier results are treated as appendix stress evidence only. Primary claims are based on medium-tier matched runs, where governance separation is measurable and non-saturated.

2 Methods

The substrate is a minimal renewable fishery commons with a shared stock, per-agent harvest actions, regeneration, and collapse dynamics. Governance mechanisms modify strategic pressure through monitoring and sanctions rather than only post-hoc accounting, so institutional choices can causally affect ecological outcomes.

Each run is evaluated under three governance settings: none, monitoring, and monitoring with sanctions. Primary contrasts are reported as the difference between monitoring with sanctions and none. For collapse readability, we also report the same comparison as "collapse reduction," where positive values indicate lower collapse under governance.

Two injector sources are matched. The first is mutation, a non-LLM strategy variation baseline. The second is live JSON-based LLM injection, where strategies are generated online by the Ollama model `qwen2.5:3b-instruct` and adapted into executable policy specifications.

The core matrix spans the easy_v1 and medium_v1 tiers and both injector types, with shared settings. Easy-tier cells use three paired runs and medium-tier high-power cells use five paired runs.

Table 1: Shared run settings for matched comparisons.

Setting	Value
Generations	15
Train seeds per generation	32
Test seeds per generation	32
Replacement fraction	0.2
Adversarial pressure	0.3
Train regeneration rate	2.0
Train observation noise std	6.0

Selection and replacement are performed in train conditions, while governance robustness is evaluated on held-out regime packs each generation. This separates adaptation pressure from generalization assessment.

The primary outcome is held-out collapse reduction. Secondary outcomes are held-out mean stock, time-to-collapse, and per-regime survival fraction over generations. A composite governance score averages paired effect sizes for collapse reduction, stock gain, and survival gain.

Analysis is run-level and confidence intervals are estimated with a bootstrap mean procedure using 5,000 resamples and percentile bounds at 2.5 and 97.5. Gate logic is evaluated from frozen summary artifacts to prevent post-hoc threshold drift. Formal invasion dynamics, the definition of collapse-robustness, and the saturation identifiability condition are given in Section 3. In the empirical pipeline, we estimate these quantities at run level and report bootstrap confidence intervals.

Methods and run metadata are version-frozen for reproducibility, including configuration manifests, gate outcomes, and the summary tables used to generate the reported statistics.

3 Formal Invasion-Stability Model

This section formalizes the invasion process evaluated in the experiments. Let $\mathcal{S}_t$ denote the strategy pool at generation t, with population distribution $p_t \in \Delta(\mathcal{S}_t)$. Let $g \in \mathcal{G}$ denote governance condition, and let $r \sim R_{\text{train}}$ or $r \sim R_{\text{test}}$ denote train or held-out regime draws.

For strategy $s \in \mathcal{S}_t$, define train-regime fitness as

$$F_t(s; g) = \mathbb{E}_{r \sim R_{\text{train}}} \left[U(s, p_t, g, r) \right],$$

where U is episodic return. Selection keeps a renormalized top-mass distribution $\mathcal{T}_q(p_t, F_t)$, where $q \in (0, 1)$ is replacement fraction. New mass is injected by generator $\mathcal{I}_i$, where i indexes injector type (mutation or live LLM). The population update is

$$p_{t+1} = (1 - q)\, \mathcal{T}_q(p_t, F_t) + q\, \mathcal{I}_i(\mathcal{H}_t),$$

with history $\mathcal{H}_t$ containing observed strategy performance up to generation t.

Let

$$C(p_t, g, r) = \mathbb{E}[\mathbf{1}\{\text{collapse}\} \mid p_t, g, r]$$

be collapse probability at generation t. For a horizon T, define invasion-robustness on collapse under governance g versus no governance as

$$\mathcal{R}^{\text{collapse}}_{g,i}(T) = \frac{1}{T} \sum_{t=1}^{T} \mathbb{E}_{r \sim R_{\text{test}}} \left[C\left(p_t^{\text{none},i}, \text{none}, r\right) - C\left(p_t^{g,i}, g, r\right) \right].$$

Positive values indicate lower collapse under governance during invasion. The paper's reported $\Delta^{\text{collapse}}_{g,i}$ is the empirical estimate of this quantity.

A practical inference rule follows directly: governance is considered invasion-stable on collapse when the lower bound of the 95% confidence interval for $\mathcal{R}^{\text{collapse}}_{g,i}(T)$ is strictly positive.

Proposition (Saturation non-identifiability). If for a regime pack R, horizon T, and all governance conditions $g \in \mathcal{G}$,

$$C\left(p_t^{g,i}, g, r\right) = 1 \quad \forall t \in \{1, \ldots, T\},\ \forall r \sim R,$$

then collapse-based governance ranking is non-identifiable because $\mathcal{R}^{\text{collapse}}_{g,i}(T) = 0$ for all g.

Sketch. Substituting $C = 1$ into the definition of $\mathcal{R}^{\text{collapse}}_{g,i}(T)$ yields zero difference at every time and regime, so all governance conditions tie under collapse-based scoring.

This proposition explains why hard-tier saturation is treated as stress confirmation rather than mechanism-separation evidence.

4 Results

Under matched settings with five paired high-power runs, monitoring with sanctions reduced held-out collapse relative to none for both injector types. For mutation, collapse reduction was 0.109896 with 95% CI $[0.031042, 0.189583]$. For live JSON-based LLM injection, collapse reduction was 0.083542 with 95% CI $[0.015417, 0.150938]$. Both confidence-interval lower bounds are positive, supporting the medium-tier robustness claim for governance under invasion pressure.

Monitoring with sanctions also improved supporting robustness indicators relative to none. Mean held-out stock increased by 23.352178 for mutation with 95% CI $[15.873344, 31.697602]$, and

by 22.402775 for live JSON-based LLM injection with 95% CI $[18.453585, 27.746404]$. Per-regime survival over generations improved by 0.109896 for mutation with 95% CI $[0.031042, 0.189583]$, and by 0.083542 for live JSON-based LLM injection with 95% CI $[0.015417, 0.151667]$. Time to collapse increased by 1.2 generations for mutation with 95% CI $[0.4, 2.2]$, and by 1.8 generations for live JSON-based LLM injection with 95% CI $[1.0, 3.0]$.

To contextualize magnitude, baseline medium-tier collapse under no governance is high: 0.9068 for mutation and 0.9267 for live JSON-based LLM injection. The observed improvements correspond to relative collapse reductions of 11.93% and 9.26%, respectively. Baseline time-to-collapse is 2.4 generations for both injectors, so governance-induced delays of 1.2 and 1.8 generations correspond to relative increases of 45.83% and 70.00%, respectively.

From the frozen gate file, the cycle-level criteria evaluate as follows: the primary medium reduction gate is PASS, the strong confidence-interval lower-bound gate is PASS, and the medium saturation check is FALSE.

Easy-tier runs used three paired runs and show wider uncertainty, especially for live JSON-based LLM injection. For collapse reduction, mutation remains positive at 0.170139 with 95% CI $[0.068750, 0.279167]$, while live JSON-based LLM injection crosses zero at 0.163542 with 95% CI $[-0.061458, 0.411979]$. This is consistent with treating medium-tier high-power evidence as the decision-critical result.

Table 2: Collapse reduction from none to monitoring with sanctions by tier and injector. Positive values indicate lower collapse under governance.

Tier	Injector	*n_pairs*	Mean reduction	95% CI
easy_v1	mutation	3	0.170139	[0.068750, 0.279167]
easy_v1	Live JSON LLM	3	0.163542	[-0.061458, 0.411979]
medium_v1	mutation	5	0.109896	[0.031042, 0.189583]
medium_v1	Live JSON LLM	5	0.083542	[0.015417, 0.150938]

Table 3 summarizes the magnitude context in medium-tier runs.

Table 3: Medium-tier magnitude context for monitoring with sanctions relative to none.

Injector	Baseline collapse	Abs. reduction	Rel. reduction	Rel. TTC gain
Mutation	0.9068	0.1099	11.93%	45.83%
Live JSON LLM	0.9267	0.0835	9.26%	70.00%

Mechanism proxies suggest that governance acts by suppressing aggressive extraction pressure while improving ecological state. In medium-tier runs, the mean high-harvest fraction drops from 0.874 to 0.790 for mutation and from 0.864 to 0.781 for live JSON-based LLM injection when moving from none to monitoring with sanctions. Over the same comparison, mean held-out stock rises from 28.4 to 51.7 for mutation and from 25.7 to 48.1 for live JSON-based LLM injection.

Using the composite effect-size score across collapse reduction, stock gain, and survival gain, live JSON-based LLM injection obtains a score of 1.805738 and mutation obtains a score of 1.490345 in the medium tier. To test whether this ranking is an artifact of equal weighting, we perform a random-weight sensitivity analysis over the same three components. Across 200,000 random weight draws, the live JSON-based LLM score ranks first in 89.99% of cases; equivalently, it ranks first whenever stock-effect weight exceeds approximately 0.0904.

Figures 1–4 visualize governance effects, time-to-collapse improvements, regime-level survival dynamics, and mechanism proxies.

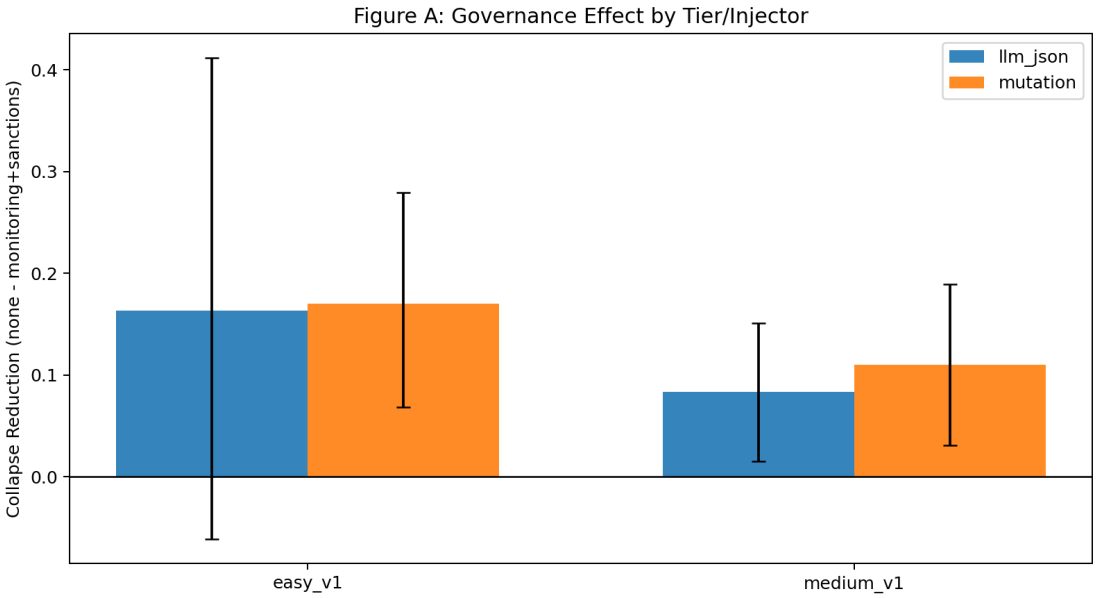

Figure 1: Governance effect by tier/injector with confidence intervals.

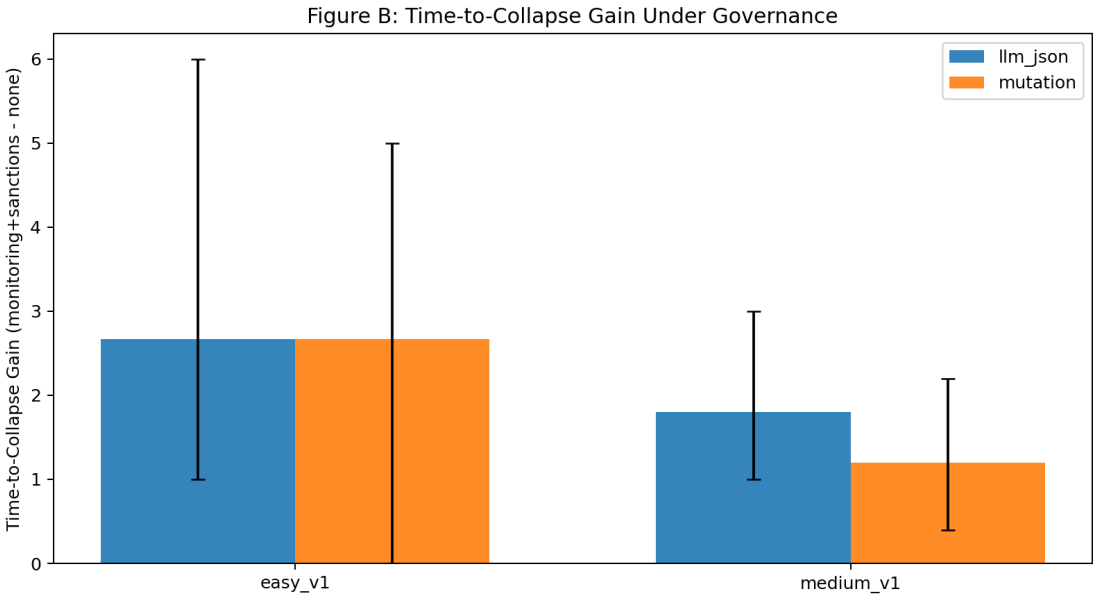

Figure 2: Time-to-collapse gain under governance.

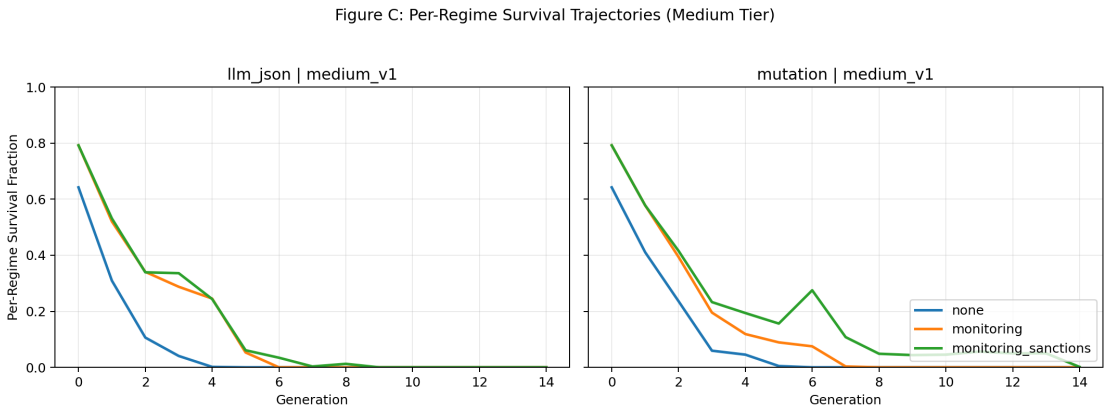

Figure 3: Per-regime survival trajectories in medium tier.

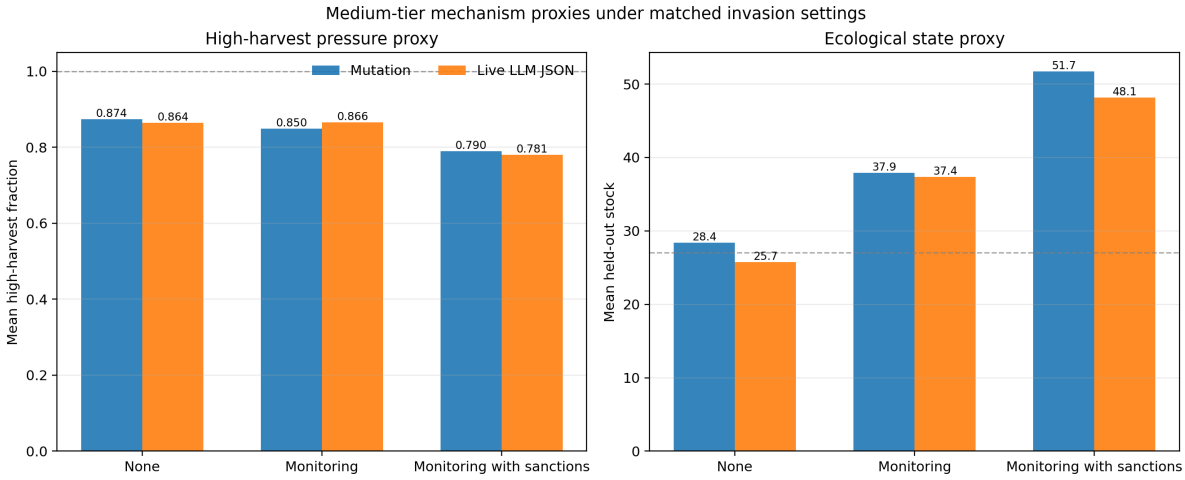

Figure 4: Medium-tier mechanism proxies with numeric labels. Left: mean high-harvest fraction by governance condition and injector. Right: mean held-out stock by governance condition and injector.

5 Discussion

The hard tier is treated as appendix-only evidence because it enters a metric-saturation regime where governance separation is no longer identifiable. In the hard-tier smoke run, all governance conditions converge to held-out collapse equal to 1.0, per-regime survival equal to 0.0, and time-to-collapse equal to 0.0. This is a collapse-floor effect: once the system reliably collapses immediately under every condition, collapse-based metrics have no remaining dynamic range for mechanism comparison. In this setting, the hard tier is still useful as a stress confirmation that the environment can reach failure extremes, but it cannot support causal ranking of governance interventions.

The environment is intentionally minimal, and this is a design choice for identification rather than a claim of full ecological realism. By restricting dynamics to a controlled renewable-stock game, the mapping from governance signals to invasion outcomes remains interpretable, and the formal quantities in Section
refsec:theory can be estimated without confounding from additional modules such as market-

price feedback, spatial migration, or multi-resource coupling. The trade-off is external validity: larger-world policy conclusions require re-testing the same invasion-stability operator after adding complexity.

The broader implication for generative societies is that the contribution is not fishery-specific, but protocol-specific. Recent LLM-agent society studies show emergence and fragility in social outcomes [7, 8], while adversarial generation work emphasizes persistent pressure and strategy turnover [9]. The present results suggest a transferable governance insight: under repeated strategy injection, lightweight institutional signals that increase observability and accountability can reduce exploitative pressure and preserve shared-system health in held-out regimes. This supports using invasion-stability evaluation as a bridge from controlled commons simulations to more complex generative-society governance settings [5].

6 Limitations

LLM runs are slower and can be sensitive to local runtime load. Confidence intervals are finite-sample and remain wider on easy tier. Hard_v1 shows saturation in smoke profile and is not used as primary mechanism-comparison evidence; once collapse saturates at 1.0 and survival at 0.0 across conditions, governance effects become non-identifiable under floor effects. The current substrate is intentionally minimal, so external validity to richer socio-ecological systems depends on future re-tests with added heterogeneity, shocks, and institutional complexity.

7 Appendix

Hard-tier stress evidence is included as appendix-only context. Detailed hard-tier run notes and artifacts are maintained as supplementary materials.

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